

Homothetic tube model predictive control with multi-step predictors

Danilo Saccani,

Giancarlo Ferrari Trecate, Melanie Zeilinger, Johannes Köhler



□ **Goal:** constrained predictive control of an unknown linear system

- Derive a model from data
- Design a controller robust against parametric uncertainty



Problem: sequential propagation of parametric uncertainty can result in conservative behaviour



Solution: change parametrization of the model to exploit reduced conservatism of multi-step prediction models and to avoid propagation [Terzi E. '18][Köhler J. '22]

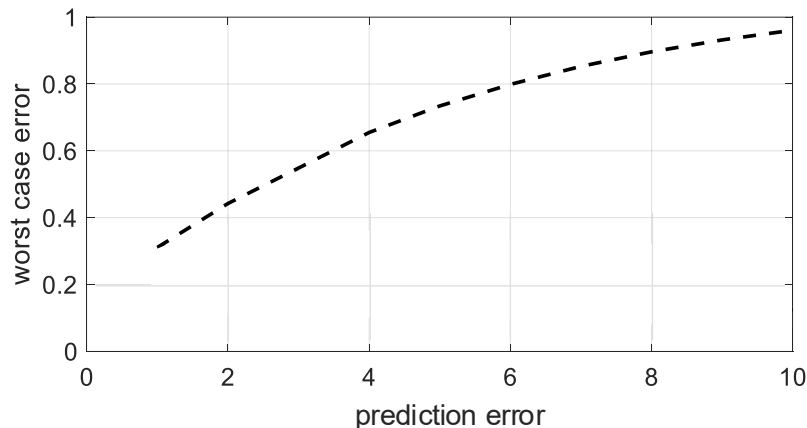


Figure adapted from [1]

[Terzi E. '18] Terzi, E., Fagiano, L., Farina, M., & Scattolini, R. (2018, June). Learning multi-step prediction models for receding horizon control. In 2018 European Control Conference (ECC) (pp. 1335-1340). IEEE.

[Köhler J '22] Köhler, J., Wabersich, K. P., Berberich, J., & Zeilinger, M. N. (2022, December). State space models vs. multi-step predictors in predictive control: Are state space models complicating safe data-driven designs?. In 2022 IEEE 61st Conference on Decision and Control (CDC) IEEE.

- 1 Multi-steps prediction model
- 2 Homothetic tube MPC formulation
- 3 Theoretical analysis
- 4 Numerical example and discussion

- Discrete time LTI system

$$x_{k+1} = Ax_k + Bu_k + Mw_k$$

Bounded disturbance
 $w_k \in \mathbb{W}$

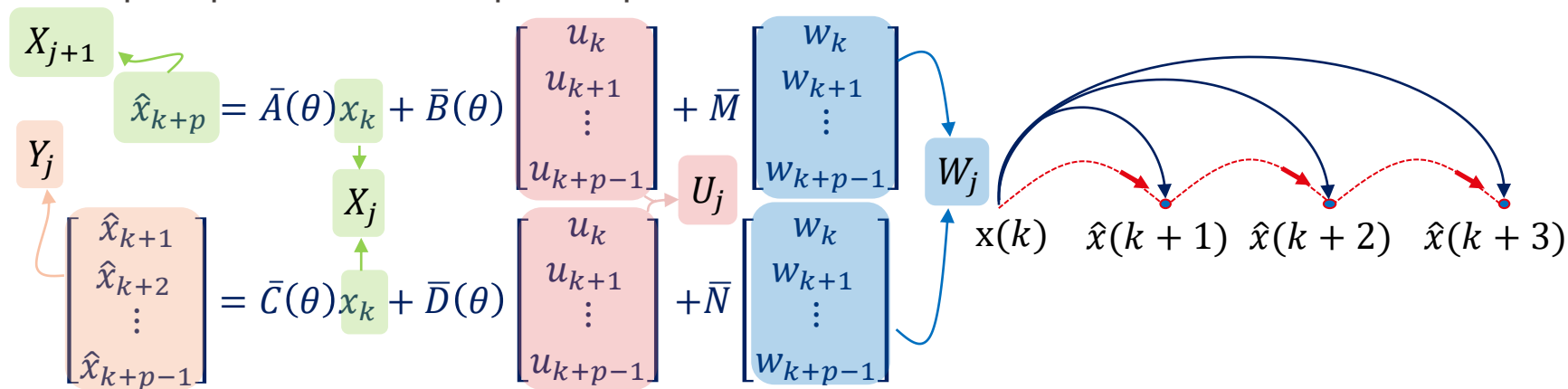
unknown matrices

- Subject to state and input constraints

$$x_k \in \mathbb{X}, \quad u_k \in \mathbb{U}$$

convex polytopes

- p-steps ahead state space representation



Assumptions:

- The matrices $\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)$ depend affinely on the parameter vector θ :

$$\left(\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)\right) = \left(\bar{A}_0, \bar{B}_0, \bar{C}_0, \bar{D}_0\right) + \sum_{i=1}^{n_p} \left(\bar{A}_i, \bar{B}_i, \bar{C}_i, \bar{D}_i\right) [\theta]_i$$

Assumptions:

- ❑ The matrices $\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)$ depend affinely on the parameter vector θ :

$$\left(\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta) \right) = \left(\bar{A}_0, \bar{B}_0, \bar{C}_0, \bar{D}_0 \right) + \sum_{i=1}^{n_p} \left(\bar{A}_i, \bar{B}_i, \bar{C}_i, \bar{D}_i \right) [\theta]_i$$

- ❑ The true parameter vector θ^* lies inside a known bounded polytopic set:

$$\Theta = \{ \theta \in \mathbb{R}^{n_p} \mid H_\theta \theta \leq h_\theta \}$$

Assumptions:

- ❑ The matrices $\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)$ depend affinely on the parameter vector θ :

$$\left(\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta) \right) = \left(\bar{A}_0, \bar{B}_0, \bar{C}_0, \bar{D}_0 \right) + \sum_{i=1}^{n_p} \left(\bar{A}_i, \bar{B}_i, \bar{C}_i, \bar{D}_i \right) [\theta]_i$$

- ❑ The true parameter vector θ^* lies inside a known bounded polytopic set:

$$\Theta = \{ \theta \in \mathbb{R}^{n_p} \mid H_\theta \theta \leq h_\theta \}$$

- ❑ The disturbance is bounded such that

$$\bar{M}W_j \in \mathbb{W}_x \qquad \bar{N}W_j \in \mathbb{W}_y$$

where $\mathbb{W}_x, \mathbb{W}_y$ are known bounded polytopic sets

Assumptions:

- ❑ The matrices $\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)$ depend affinely on the parameter vector θ :

$$(\bar{A}(\theta), \bar{B}(\theta), \bar{C}(\theta), \bar{D}(\theta)) = (\bar{A}_0, \bar{B}_0, \bar{C}_0, \bar{D}_0) + \sum_{i=1}^{n_p} (\bar{A}_i, \bar{B}_i, \bar{C}_i, \bar{D}_i) [\theta]_i$$

- ❑ The true parameter vector θ^* lies inside a known bounded polytopic set:

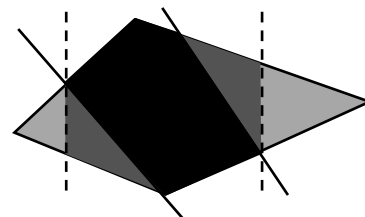
$$\Theta = \{\theta \in \mathbb{R}^{n_p} \mid H_\theta \theta \leq h_\theta\}$$

- ❑ The disturbance is bounded such that

$$\bar{M}W_j \in \mathbb{W}_x \quad \bar{N}W_j \in \mathbb{W}_y$$

where $\mathbb{W}_x, \mathbb{W}_y$ are known bounded polytopic sets

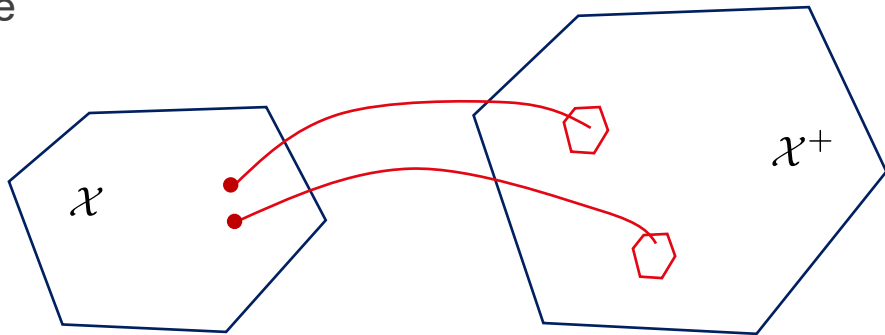
set membership identification allows to obtain the desired bounds directly from data [Lauricella M. '20]



- 1 Multi-steps prediction model
- 2 Homothetic tube MPC formulation
- 3 Theoretical analysis
- 4 Numerical example and discussion

- Uncertain state predictions lie in tube

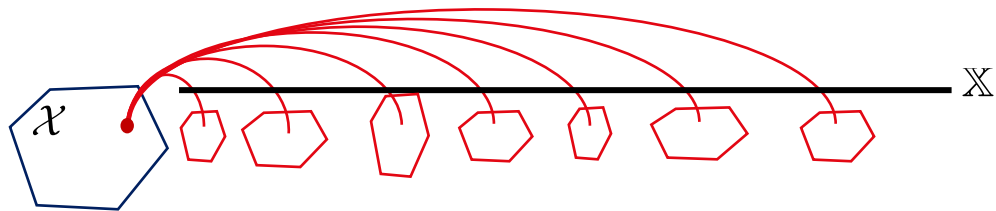
$$X^+ \in \mathcal{X}^+, \forall X \in \mathcal{X}, \\ \forall \theta \in \Theta, \forall \bar{M}W \in \mathbb{W}_x$$



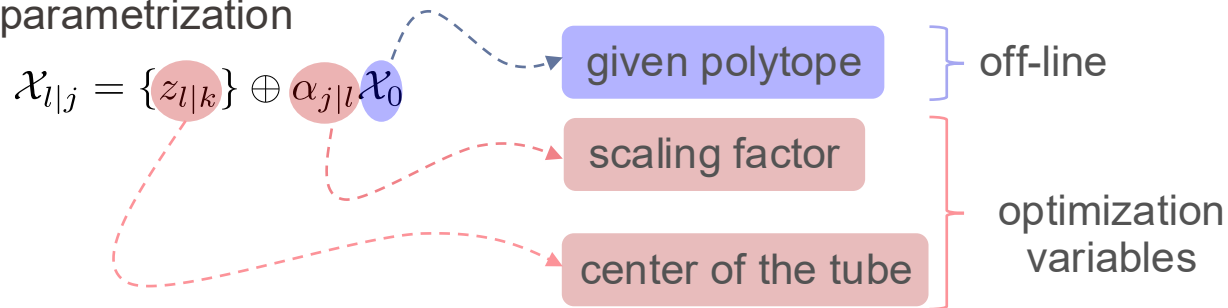
- State and input constraints are satisfied at each time step

$$Y \in \mathbb{X}^{p-1} \quad U(X) \in \mathbb{U}^p \\ \forall \theta \in \Theta, \forall \bar{N}W \in \mathbb{W}_y, \forall X \in \mathcal{X}$$

$$Y = [x_{k+1}^\top, \dots, x_{k+p-1}^\top]^\top$$



- Tube parametrization



Proposition: Linear parametrization of the tube

$$\begin{aligned}
 &U(X) \in \mathbb{U}^p, \forall X \in \mathcal{X}, \\
 &X^+ \in \mathcal{X}^+, Y \in \mathbb{X}^{p-1}, \\
 &\forall \theta \in \Theta, \forall X \in \mathcal{X} \\
 &\forall \bar{M}W \in \mathbb{W}_x, \forall \bar{N}W \in \mathbb{W}_y
 \end{aligned}$$

are satisfied if and only if
there exists $\Lambda_{l|j}^v \geq 0$

such that

$$\begin{aligned}
 &Fz + \alpha \bar{f} \leq \mathbf{1}, \\
 &G_p Kz + G_p V + \alpha \bar{g} \leq \mathbf{1}, \\
 &\Lambda^v h_\theta + H_p e^v \leq \Phi^+ - \bar{w}, \\
 &H_p E^v = \Lambda^v H_\theta, \forall v \in \mathbb{N}_1^{n_v}
 \end{aligned}$$

$$\min \sum_{l=0}^{N_p-1} (\|\hat{Y}_{l|j}\|_{Q_p}^2 + \|\hat{X}_{l|j}\|_Q^2 + \|\hat{U}_{l|j}\|_{R_p}^2) + \|\hat{X}_{N_p|j}\|_P^2$$

$$\hat{X}_{l+1|j} = \bar{A}(\hat{\theta})\hat{X}_{l|j} + \bar{B}(\hat{\theta})\hat{U}_{l|j}, \quad \hat{Y}_{l|j} = \bar{C}(\hat{\theta})\hat{X}_{l|j} + \bar{D}(\hat{\theta})\hat{U}_{l|j},$$

$$\hat{X}_{0|j} = X_j, \quad \hat{U}_{l|j} = K\hat{X}_{l|j} + V_{l|j}, \quad \forall l \in \mathbb{N}_0^{N_p-1}$$

Certainty equivalent
regulation cost

$$\mathcal{X}_{0|j} \ni X_j$$

$$X^+ \in \mathcal{X}_{l+1|j}, \quad \forall \bar{M}W \in \mathbb{W}_x, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$x \in \mathbb{X}, \quad U_{l|j}(X) \in \mathbb{U}^p, \quad \forall X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$Y \in \mathbb{Y}^{p-1}, \quad \forall \bar{N}W \in \mathbb{W}_y, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$z_{N_p|j} \oplus \alpha_{N_p|j} \mathcal{X}_0 \subseteq \mathcal{X}_f$$

$$\min \sum_{l=0}^{N_p-1} (\|\hat{Y}_{l|j}\|_{Q_p}^2 + \|\hat{X}_{l|j}\|_Q^2 + \|\hat{U}_{l|j}\|_{R_p}^2) + \|\hat{X}_{N_p|j}\|_P^2$$

$$\hat{X}_{l+1|j} = \bar{A}(\hat{\theta})\hat{X}_{l|j} + \bar{B}(\hat{\theta})\hat{U}_{l|j}, \quad \hat{Y}_{l|j} = \bar{C}(\hat{\theta})\hat{X}_{l|j} + \bar{D}(\hat{\theta})\hat{U}_{l|j},$$

$$\hat{X}_{0|j} = X_j, \quad \hat{U}_{l|j} = K\hat{X}_{l|j} + V_{l|j}, \quad \forall l \in \mathbb{N}_0^{N_p-1}$$

$$\mathcal{X}_{0|j} \ni X_j$$

$$X^+ \in \mathcal{X}_{l+1|j}, \quad \forall \bar{M}W \in \mathbb{W}_x, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$x \in \mathbb{X}, \quad U_{l|j}(X) \in \mathbb{U}^p, \quad \forall X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$Y \in \mathbb{Y}^{p-1}, \quad \forall \bar{N}W \in \mathbb{W}_y, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$z_{N_p|j} \oplus \alpha_{N_p|j} \mathcal{X}_0 \subseteq \mathcal{X}_f$$

Linear constraints for the
homothetic tube

$$\min \sum_{l=0}^{N_p-1} (\|\hat{Y}_{l|j}\|_{Q_p}^2 + \|\hat{X}_{l|j}\|_Q^2 + \|\hat{U}_{l|j}\|_{R_p}^2) + \|\hat{X}_{N_p|j}\|_P^2$$

$$\hat{X}_{l+1|j} = \bar{A}(\hat{\theta})\hat{X}_{l|j} + \bar{B}(\hat{\theta})\hat{U}_{l|j}, \quad \hat{Y}_{l|j} = \bar{C}(\hat{\theta})\hat{X}_{l|j} + \bar{D}(\hat{\theta})\hat{U}_{l|j},$$

$$\hat{X}_{0|j} = X_j, \quad \hat{U}_{l|j} = K\hat{X}_{l|j} + V_{l|j}, \quad \forall l \in \mathbb{N}_0^{N_p-1}$$

$$\mathcal{X}_{0|j} \ni X_j$$

$$X^+ \in \mathcal{X}_{l+1|j}, \quad \forall \bar{M}W \in \mathbb{W}_x, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$x \in \mathbb{X}, \quad U_{l|j}(X) \in \mathbb{U}^p, \quad \forall X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$Y \in \mathbb{Y}^{p-1}, \quad \forall \bar{N}W \in \mathbb{W}_y, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$z_{N_p|j} \oplus \alpha_{N_p|j} \mathcal{X}_0 \subseteq \mathcal{X}_f$$

Terminal set and terminal cost such that $\mathcal{X}_f = \eta \mathcal{X}_0$ is an RPI set for $X_{j+1} = \bar{A}(\theta)X_j + \bar{B}(\theta)U_j$ and P is such that

$$\bar{A}_{cl}(\theta)^\top P \bar{A}_{cl}(\theta) + Q + \bar{C}_{cl}(\theta)^\top Q_p \bar{C}_{cl}(\theta) + K^\top R_p K \preceq P \quad \forall \theta \in \Theta.$$

$$\min \sum_{l=0}^{N_p-1} (\|\hat{Y}_{l|j}\|_{Q_p}^2 + \|\hat{X}_{l|j}\|_Q^2 + \|\hat{U}_{l|j}\|_{R_p}^2) + \|\hat{X}_{N_p|j}\|_P^2$$

$$\hat{X}_{l+1|j} = \bar{A}(\hat{\theta})\hat{X}_{l|j} + \bar{B}(\hat{\theta})\hat{U}_{l|j}, \quad \hat{Y}_{l|j} = \bar{C}(\hat{\theta})\hat{X}_{l|j} + \bar{D}(\hat{\theta})\hat{U}_{l|j},$$

$$\hat{X}_{0|j} = X_j, \quad \hat{U}_{l|j} = K\hat{X}_{l|j} + V_{l|j}, \quad \forall l \in \mathbb{N}_0^{N_p-1}$$

$$\mathcal{X}_{0|j} \ni X_j$$

$$X^+ \in \mathcal{X}_{l+1|j}, \quad \forall \bar{M}W \in \mathbb{W}_x, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$x \in \mathbb{X}, \quad U_{l|j}(X) \in \mathbb{U}^p, \quad \forall X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$Y \in \mathbb{Y}^{p-1}, \quad \forall \bar{N}W \in \mathbb{W}_y, \quad \theta \in \Theta, \quad X \in \mathcal{X}_{l|j}, \quad l \in \mathbb{N}_0^{N_p-1}$$

$$z_{N_p|j} \oplus \alpha_{N_p|j} \mathcal{X}_0 \subseteq \mathcal{X}_f$$



$$U_j = \begin{bmatrix} u_k \\ u_{k+1} \\ \vdots \\ u_{k+p-1} \end{bmatrix}$$

The optimization problem is solved every p -steps, and apply feedback $U(X)$

- 1 Multi-steps prediction model
- 2 Homothetic tube MPC formulation
- 3 Theoretical analysis
- 4 Numerical example and discussion

Theorem:

Let the assumptions hold, and suppose the a terminal set is properly defined. If the problem is feasible at time $j = 0$, then, the MPC scheme is feasible for all time instant $j > 0$, the resulting closed-loop system robustly satisfies constraints for all $k \in \mathbb{N}$ and $X_j = 0$ is practically asymptotically stable.

Theorem:

Let the assumptions hold, and suppose the a terminal set is properly defined. If the problem is feasible at time $j = 0$, then, the MPC scheme is feasible for all time instant $j > 0$, the resulting closed-loop system robustly satisfies constraints for all $k \in \mathbb{N}$ and $X_j = 0$ is practically asymptotically stable.

□ Proof (sketch)

Recursive feasibility: standard shifted sequences arguments.

Practical stability: standard arguments from robust MPC

- Bound of the value function variation in case of exact predictions.
- Since the value function of the QP is piecewise quadratic and the prediction error is bounded there exist a constant $\epsilon > 0$, such that:

$$J_N^*(X_{j+1}) \leq J_N^*(X_j) - \|X_j\|_Q^2 + \gamma(\epsilon)$$

- Finally $J_N^*(X)$ is piece-wise quadratic ensuring that it is lower and upper bounded by a $\mathcal{K}_\infty$ function of $\|X\|$
→ Thus, $J_N^*(X)$ is a practical Lyapunov function.

- 1 Multi-steps prediction model
- 2 Homothetic tube MPC formulation
- 3 Theoretical analysis
- 4 Numerical example and discussion

- Discretized third order system identified with multi-step horizon $p = 10$

$$G(s) = \frac{160}{(s + 10)(s^2 + 1.6s + 16)}$$

- Objective: minimize x_3
- State and input constraints $\mathbb{X} = \{x \in [-10, 10]^3, x_3 \geq -1\}$, $\mathbb{U} = [-10, 10]$
- Comparison in terms of conservatism and complexity with:

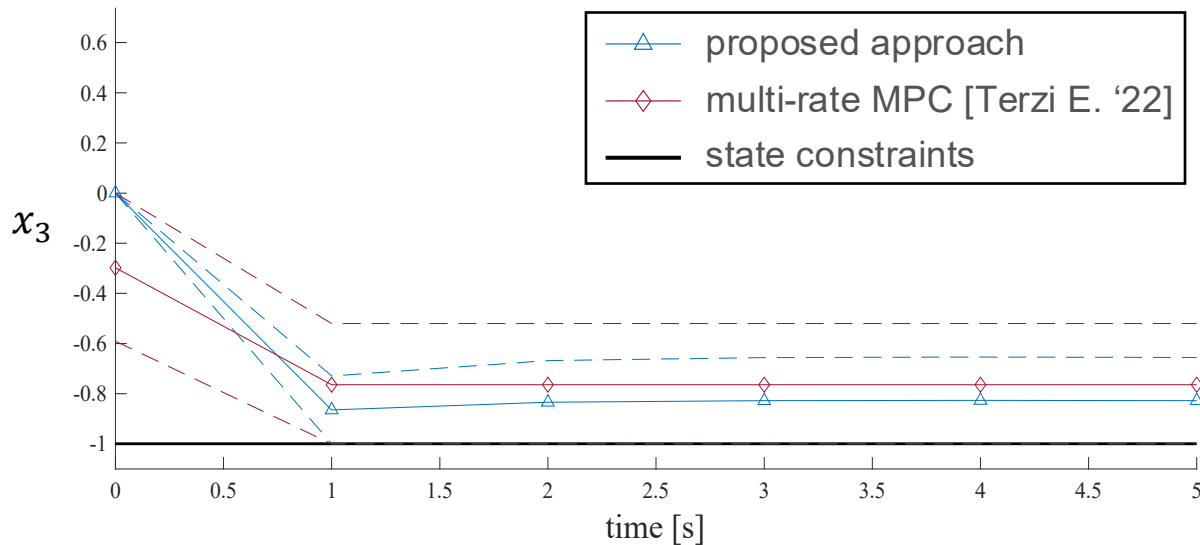


[Terzi E. '22] E. Terzi, M. Farina, L. Fagiano, and R. Scattolini, "Robust multi-rate predictive control using multi-step prediction models learned from data," Automatica, 2022.

[Lorenzen M. '19] M. Lorenzen, M. Cannon, and F. Allgöwer, "Robust MPC with recursive model update," Automatica, 2019.

Conservatism:

- multi-rate MPC [Terzi E. '22] utilizes a constant upper bound on the prediction error for constraints tightening and construction RPI set.
- proposed homothetic tube formulation non-conservatively accounts for the parametric uncertainty $\theta \in \Theta$.



[Terzi E. '22] E. Terzi, M. Farina, L. Fagiano, and R. Scattolini, "Robust multi-rate predictive control using multi-step prediction models learned from data," Automatica, 2022.

□ Complexity:

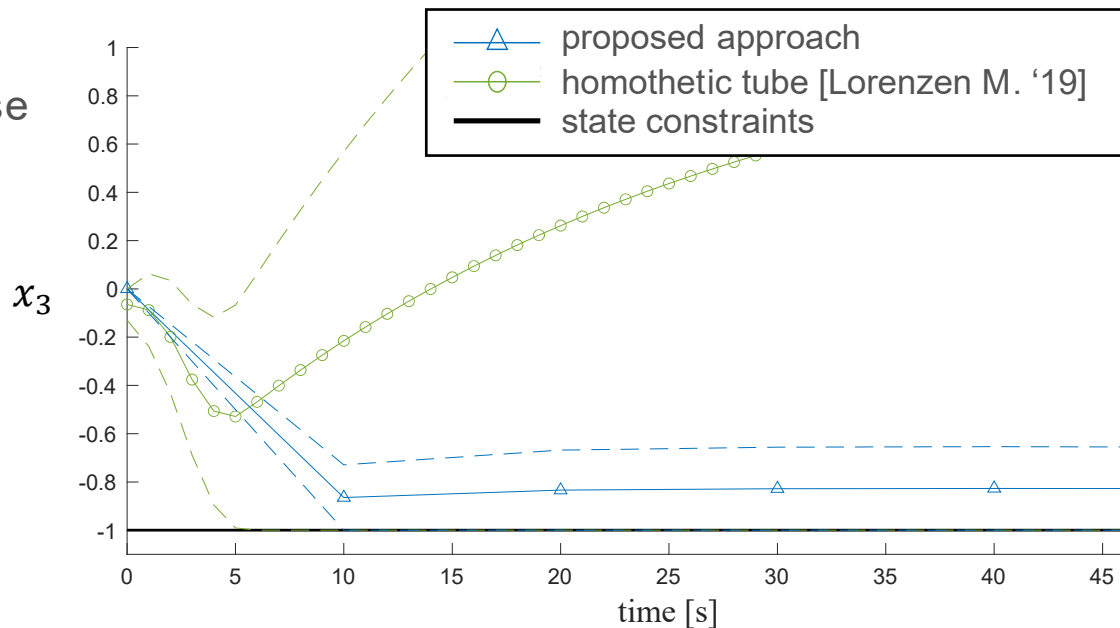
- Complexity multi-rate MPC [Terzi E. '22] similar to nominal MPC.
- homothetic MPC increased number of optimization variables due to tube parametrization

Approach	#opt. var.	#ineq. con.	#eq. con.
multi-rate MPC [Terzi E. '22]	$2.5 \cdot 10^2$	$9.5 \cdot 10^2$	$1.5 \cdot 10^2$
proposed approach	$1.2 \cdot 10^5$	$1.2 \cdot 10^5$	$6.1 \cdot 10^5$

$$\begin{aligned}
 Fz_{l|j} + \alpha_{l|j}\bar{f} &\leq \mathbf{1}, \\
 G_p K z_{l|j} + G_p V_{l|j} + \alpha_{l|j}\bar{g} &\leq \mathbf{1}, \\
 H_x(x_k - z_{0|j}) &\leq \alpha_{0|j}\mathbf{1}, \\
 \Lambda_{l|j}^v h_\theta + H_p e_{l|j}^v &\leq \Phi_{l+1|j} - \bar{w}, \\
 H_p E_{l|j}^v &= \Lambda_{l|j}^v H_\theta,
 \end{aligned}$$

□ Conservatism:

- In [Lorenzen M. '19], each time step requires a tube propagation that results in conservatism due to the fixed parametrization of $\mathcal{X}_0$.
In contrast, the proposed formulation requires $1/p$ such over-approximations.
- [Lorenzen M. '19] more conservative due to the use of the same parametrized tube $\mathcal{X}_0$.



□ Complexity:

- Complexity in [Lorenzen M. '19] similar to the proposed approach with the same $\mathcal{X}_0$.

Approach	#opt. var.	#ineq. con.	#eq. con.
homothetic tube [Lorenzen M. '19]	$7.1 \cdot 10^4$	$7.2 \cdot 10^4$	$3.4 \cdot 10^4$
proposed approach	$1.2 \cdot 10^5$	$1.2 \cdot 10^5$	$6.1 \cdot 10^5$

- We can use low complexity polytopes (reduced number of vertices), thus reducing the overall complexity.
- The proposed approach solves the optimization problem every p time instants.

MPC scheme with multi-step predictors that combines:

- multi-rate MPC formulation ➤ reduces the complexity of homothetic tube MPC with state space models.
- homothetic tubes ➤ reduces the conservatism of fixed tube approaches.

Future works

- Natural extension to control scheme with adaptive model update.
- Address scalability limitations associated with vertex enumeration by utilizing ellipsoids.

Thank you!



□ Complexity:

Approach	#opt. var.	#ineq. con.	#eq. con.
Multi-rate MPC [25]	$2.5 \cdot 10^2$	$9.5 \cdot 10^2$	$1.5 \cdot 10^2$
Homothetic tube [6]	$7.1 \cdot 10^4$	$7.2 \cdot 10^4$	$3.4 \cdot 10^4$
Homothetic tube - V2 [7]	$3.0 \cdot 10^2$	$1.2 \cdot 10^6$	$2.0 \cdot 10^2$
Proposed (20)	$1.2 \cdot 10^5$	$1.2 \cdot 10^5$	$6.1 \cdot 10^5$

TABLE I

COMPARISON OF COMPUTATIONAL COMPLEXITY OF THE QPs.

[25] E. Terzi, M. Farina, L. Fagiano, and R. Scattolini, "Robust multi-rate predictive control using multi-step prediction models learned from data," *Automatica*, 2022.

[6] M. Lorenzen, M. Cannon, and F. Allgöwer, "Robust MPC with recursive model update," *Automatica*, 2019.

[7] J. Köhler, E. Andina, R. Soloperto, M. A. Müller, and F. Allgöwer, "Linear robust adaptive model predictive control: Computational complexity and conservatism," in *Proc. 58th IEEE Conf. Decision and Control (CDC)*, 2019, pp. 1383–1388